

Simulated Observation of the de Broglie Wavelength in a Localized Odd-Harmonic Double-Slit Model

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— *Interference-survival condition of a highly localized structure, and the fundamental-wavelength invariance of the observed wavelength*

1 Abstract

The de Broglie wavelength $\lambda = h/p$ was first confirmed by the 1927 Davisson–Germer electron diffraction [2]. This paper is a numerical experiment that **simulates and observes** this relation inside a **single-source double-slit classical wave model** (the interference and alignment judgement are deterministic; only the input position and wavelength fluctuations are supplied by independent Monte Carlo draws). The electron is modeled as a **spatially localized wave** with representative wavelength $\lambda_0 = h/p$, composed of an equal-amplitude sum of odd harmonics for $N > 1$ [1] (as noted below, this is not the electron wave packet itself but a classical analogue: the minimal Fourier construction of a spatially localized pulse). Computing the exact two-slit interference at the physical scale ($L = 5\text{ cm}$, $W = 5\text{ }\mu\text{m}$): for $N = 1$ the wave always interferes and the fringe spacing recovers λ_0 (a check of the procedure). For $N > 1$, preserving the localized shape requires an inter-harmonic alignment condition (the geometric resonance $r_{\text{avg}}/\lambda = M/2$); at wavelengths that do not satisfy it, the localization collapses and no stable fringes form. Computing each electron exactly under a symmetric fluctuation centered at $\lambda_0 = h/p$, the effective wavelength λ' that interferes stably is selected as the nearest resonance. **The central result of this paper is not the value of h .** The resonance-lattice spacing is $\sim 10^{-19}\text{ m}$ —so dense that $\lambda' \approx \lambda_0$ (relative $\sim 10^{-9}$), and $p\lambda' \approx h$ is a nearly tautological self-consistency. The center of gravity of this paper is the **interference survival of the structure and the fundamental-wavelength invariance of the observed wavelength**: even when finer localized structure ($N > 1$ odd harmonics) than λ_0 is packed in, the wave can “**survive**” two-slit interference as a sharp single peak, and the observed central fringe spacing remains at the fundamental wavelength λ_0 (N controls only the sharpness of localization $\sim 1/(N+1)$). This fundamental-wavelength invariance is a **structural** result, independent of whatever physical origin the fundamental wavelength λ_0 is given; it is a claim distinct from the numerical recovery of h . **This paper is neither a first-principles derivation of $p\lambda = h$ nor a derivation of a $\Delta x \Delta p \sim h$ type uncertainty relation.** As a guard against misreading, each electron is given an independent Monte Carlo draw of its source position ($\cos^2$, $\pm\lambda_0/2$) and its initial wavelength ($\cos^2$, $\pm 1\%$), and the interference, alignment, and exclusion are recomputed exactly at each (y, λ'_0) . Under this symmetric fluctuation centered at $\lambda_0 = h/p$, $\langle p\lambda' \rangle$ agrees with h within the standard error (sample standard deviation $\sigma \approx 0.36\% =$ the theoretical value of the $\cos^2$ distribution, $\text{SE} \approx \sigma/\sqrt{n_{\text{OK}}}$); a few percent of trials whose λ'_0 falls near a resonance midpoint are excluded by the extraction procedure.

2 1. Introduction and scope

In 1924–1925 de Broglie associated a wavelength $\lambda = h/p$ with a particle [3]. This relation was confirmed by the Davisson–Germer nickel electron diffraction [2] (Bragg condition $n\lambda = 2d \sin \theta$: with known d and measured θ , λ is obtained and compared with h/p).

Scope of this paper:

- **Not** a proposal of a new measurement method, nor a re-measurement of a real experiment.
- **Not** a first-principles derivation of $p\lambda = h$ (§5.1, §5.2).
- It is a **numerical experiment in classical wave optics** borrowing the two-slit geometry, asking: what alignment condition does a **highly localized** classical wave ($N > 1$ odd harmonics) with a de Broglie-scale representative wavelength require to survive stably as sharp interference under two-slit geometry, and how is the observed wavelength scale preserved.

Davisson–Germer used a crystal “lattice”, whereas this model uses a double slit. But whereas the real experiment supplies λ independently via the **external measured datum** of the diffraction angle, this model generates fringes from $\lambda_0 = h/p$ and reads them back, so it does not have an independent measurement path and is not isomorphic in that sense (§5.3). The novelty of this paper is: applying the localized odd-harmonic / alignment condition of [1] to **the double-slit geometry**, performing **per-electron exact alignment/exclusion judgement** under **independent Monte Carlo** of position and wavelength, and statistically verifying the **fundamental-wavelength invariance** of the observed wavelength.

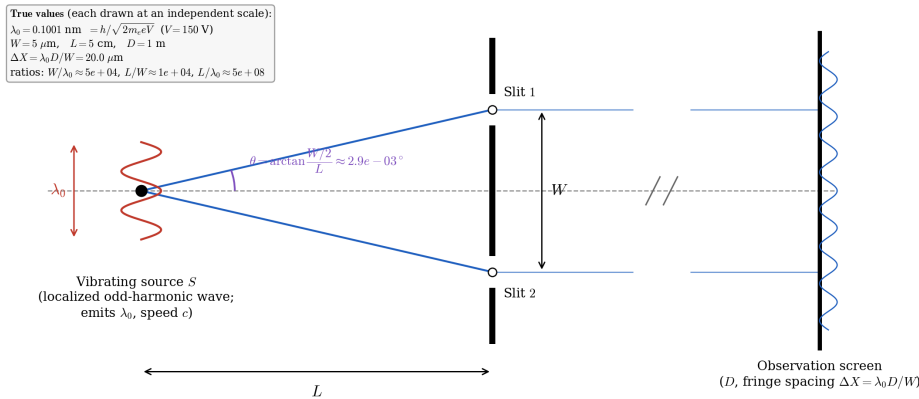
3 2. Model definition and computation

3.1 2.1 Geometry

Source–slit distance $L = 5$ cm, slit separation $W = 5 \mu\text{m}$, slits at $y_{\text{slit}} = \pm W/2$. For a source position y (which fluctuates, below) the two path lengths are

$$r_1(y) = \sqrt{L^2 + (y - W/2)^2}, \quad r_2(y) = \sqrt{L^2 + (y + W/2)^2}, \quad r_{\text{avg}} = \frac{r_1 + r_2}{2}.$$

At the centred source $y = 0$, $r_1 = r_2 = r_{\text{avg}} = \sqrt{L^2 + (W/2)^2}$. At the physical scale the characteristic lengths differ by orders of magnitude ($W/\lambda_0 \sim 5 \times 10^4$, $L/W \sim 10^4$, $L/\lambda_0 \sim 5 \times 10^8$), making a common-scale drawing impossible, and the half-angle $\theta = \arctan[(W/2)/L] \approx 2.9 \times 10^{-3}^\circ$ is extremely paraxial.



de Broglie two-slit configuration (physical scale). A vibrating source S emits the localized odd-harmonic wave of (rough) wavelength $\lambda_0 = h/\sqrt{2m_e eV}$; two slits separated by W sit at distance L ; the screen is at D . **Not to scale:** λ_0 , W and L are each drawn at an INDEPENDENT visual scale because $W/\lambda_0 \sim 5 \times 10^4$ and $L/W \sim 10^4$ make a common scale unreadable; the true magnitudes are annotated. Only the topology source $\rightarrow$ slits $\rightarrow$ screen and the paraxiality ($\theta \sim 10^{-3}^\circ$) are physical.

Figure 1: Schematic

of the setup ($V = 150 \text{ V}$). λ_0, W, L are each drawn at an independent scale (the top-left box notes the true ratios $W/\lambda_0 \approx 5 \times 10^4$, etc.). Only the topology and the paraxiality $\theta \sim 10^{-3}^\circ$ are physical; the size ratios are exaggerated (§2.1).

3.2 2.2 The localized odd-harmonic wave and the exact intensity

Physical motivation: for convenience we call it the “electron” below, but what we compute is not the full quantum state of an electron; it is a trial of a classical localized wave whose representative wavelength is at the de Broglie scale. This model treats that localized wave as a **spatially localized pulse wave**, not a point without spatial extent. An equal-amplitude sum of odd harmonics is the minimal Fourier construction that gives a symmetric localized peak at the center $s = 0$ [1], and excluding the even components preserves the symmetry. **Note:** these odd-harmonic components do **not** mean the different momentum eigenstates of a free electron in standard quantum mechanics (a factor- n wavenumber would mean momentum np and energy $n^2 E$); they are **Fourier components in a classical wave model used to construct the localized shape**. Moreover, this paper is not a time-evolving quantum wave-packet model but a **stationary classical wave model that inspects the spatial phase structure on the screen**; the energy dispersion of the harmonic components and temporal decoherence are out of scope.

Wavenumber components $n = 1, 3, \dots, N$ ($\lambda_n = \lambda/n$, $K = (N+1)/2$ of them). The exact screen intensity after the two slits is

$$I(s; y) = \left| \sum_{n=1,3,\dots,N} (e^{i\Phi_1^{(n)}} + e^{i\Phi_2^{(n)}}) \right|^2, \quad \Phi_k^{(n)} = \frac{2\pi n}{\lambda} (r_k - y_{\text{slit},k} s),$$

where s is the paraxial screen variable ($s \simeq X/D$ for the far-field coordinate X). Numerically we use the algebraically equivalent factored form

$$e^{i\Phi_1^{(n)}} + e^{i\Phi_2^{(n)}} = e^{i2\pi n r_{\text{avg}}/\lambda} \cdot 2 \cos\left(\pi n \frac{(r_1 - r_2) - Ws}{\lambda}\right), \quad r_1 - r_2 = \frac{-2Wy}{r_1 + r_2}$$

(the physics is unchanged; this avoids cancellation at absolute phases $\sim 10^9$ rad). **In this form, $e^{i2\pi n r_{\text{avg}}/\lambda}$ carries the inter-harmonic absolute alignment, while $\cos(\pi n[(r_1 - r_2) - Ws]/\lambda)$ carries the relative phase of the two-path interference.** That is, the dominant factor observed in an ordinary double slit is the path difference (relative phase), but in this model an **additional harmonic alignment with respect to the absolute mean distance r_{avg}** is required to preserve the localized odd-harmonic shape at the center.

3.3 2.3 The central alignment $A(\lambda)$ (absolute measure)

The intensity at the screen center $s = 0$ is $I(0) = |2 \sum_n e^{i2\pi n r_{\text{avg}}/\lambda}|^2 = 4K^2 A(\lambda)$,

$$A(\lambda) = \frac{1}{K^2} \left| \sum_{n=1,3,\dots,N} e^{i2\pi n r_{\text{avg}}/\lambda} \right|^2.$$

A is the **fraction of full alignment**: $A = 1$ ($I(0) = 4K^2$) only at a true resonance ($r_{\text{avg}}/\lambda = M/2$) where all odd-harmonic phases align, and $A < 1$ otherwise. As noted, A is the measure of the above (absolute) alignment, i.e. it measures **the preservation condition of the localized shape**. Because A is referred to the absolute maximum $4K^2$ fixed by geometry and N , a degraded partially-aligned pattern gives $A \ll 1$ and is correctly rejected (a self-normalized $I(0)/I_{\text{max}}$ cannot make this distinction).

3.4 2.4 Independent per-electron fluctuations and exact computation

To emulate real apparatus fluctuations, **each electron (trial) is given two independent Monte Carlo draws:**

1. **Source position** $y \sim \cos^2(\pi y/\Delta)$, range $\pm\Delta/2$ ($\Delta = \lambda_0$, peaked at the center 0).
2. **Initial wavelength** $\lambda'_0 = \lambda_0 + \eta$, $\eta \sim \cos^2(\pi\eta/2\delta)$, $\delta = 0.01 \lambda_0$ ($\pm 1\%$, drawn from a random stream independent of position).

The two draws are independent, hence statistically uncorrelated (no spurious correlation between position and wavelength). For each electron, using the geometry $r_{\text{avg}}(y)$ of its (y, λ'_0) , the interference, alignment, and exclusion are computed **exactly every time** (no $y = 0$ -fixed or λ_0 -fixed approximation; for $N = 9$ this is about 4×10^4 evaluations of A per voltage point). The theoretical standard deviation of the $\cos^2$ distribution is $\sigma = \delta\sqrt{1/3 - 2/\pi^2} \approx 0.362\delta$, i.e. $\sigma \approx 0.362\%$ for a $\pm 1\%$ input.

3.5 2.5 Search for the effective wavelength λ' and the exclusion judgement

For each electron, within the **geometric window** $[\lambda'_0 - \lambda_0'^2/(4r_{\text{avg}}), \lambda'_0 + \lambda_0'^2/(4r_{\text{avg}})]$ (= half the resonance spacing) around λ'_0 , sweep $A(\lambda)$ and count the connected bands with $A \geq 0.98$ (resonance bands). If there is exactly one band, refine its peak by golden-section search to obtain the effective wavelength $\lambda' = 2r_{\text{avg}}/M$ (satisfying $A(\lambda') = 1$, $I(0) = 4K^2$ to machine precision). If there are zero or two-or-more bands (near a resonance midpoint, the nearest resonance is not unique for λ'_0), that trial is excluded. The window width and the alignment condition are determined by geometry alone; h enters only through the center $\lambda_0 = h/p$ of λ'_0 . The observed order $M = \text{round}(2r_{\text{avg}}/\lambda')$ is a post-hoc output (the symbol M for the order is distinguished from the electron mass m_e).

Physical interpretation of the “alignment (snap)” (important): the alignment in this model does **not** claim a mechanism by which the source physically changes its wavelength after emission (e.g. injection locking via backscatter from the slits). It is an operation that **mathematically stands in for the survival bias (filtering) that “only wavelengths that align form a stable single-peak fringe, while the rest form no fringe”**. The exclusion (the “band $\neq 1$ ” trials of the previous paragraph) is a part of this filter (removal of trials for which the nearest resonance cannot be uniquely chosen); it is a **procedural judgement dependent on the threshold 0.98 and the window width**, and does not mean the electron physically disappears (the rate is set by the threshold, not by nature). By “survival” in this paper we mean that a stable single-peak fringe is extractable, not the physical life or death of an electron.

3.6 2.6 The role of λ_0 and the positions of $N = 1$ / $N > 1$

$\lambda_0 = h/p$ is the initial condition of the interference computation; without λ_0 the phase $2\pi r/\lambda$ cannot be defined. Since λ_0 itself is given using h , the recovery of $p\lambda'$ is not an independent derivation of h but a self-consistency check.

- $N = 1$ (**single harmonic**) always interferes without an alignment condition, and the same λ_0 is recovered from the fringe spacing. Hence $N = 1$ is not a physical discovery but a **check of the computation and the wavelength-readout procedure**.
- $N > 1$ (**localized odd-harmonic wave**) is the main subject. This waveform has structure finer than λ_0 , and it can interfere stably only on the discrete resonances $r_{\text{avg}}/\lambda = M/2$ (the absolute alignment condition of §2.2). Since the electron’s λ_0 generally does not lie on this lattice, it otherwise collapses, and the λ' of maximal A (the nearest resonance) is selected by the survival filter.

At the physical scale the resonance-lattice spacing $\lambda_0^2/(2r_{\text{avg}}) \sim 10^{-19}$ m is extremely dense, so λ' coincides with λ_0 to relative $\sim 10^{-9}$. Thus the “micro-adjustment of λ' ” itself is a tautology with no new physics. **The content specific to $N > 1$ is not the recovery of a value, but the following two points:** (i) a classical wave with structure finer than λ_0 can survive stably as a sharp single-peak interference under two-slit geometry, and (ii) the central spacing of the surviving fringes is at the fundamental wavelength λ_0 , independent of the harmonic count N (N controls only the sharpness $\sim 1/(N+1)$). These are structural results that do not depend on whatever physical origin λ_0 is given, and are distinct claims from the numerical recovery of h .

4 3. $N = 1$: baseline (a check of the procedure)

$N = 1$ interferes unconditionally. At each voltage $V = 50\text{--}500$ V ($\lambda_0 = 0.055\text{--}0.173$ nm), we accumulate the exact two-slit fringes of an ensemble of electrons with position fluctuation $\pm\lambda_0/2$ and wavelength fluctuation $\pm 1\%$ (Figure 2), and recover the wavelength from the fringe spacing. The slope of $\langle p\lambda \rangle$ agrees with h at slope/ $h = 0.99977$ (a statistical residual $\sim 2 \times 10^{-4}$ from averaging the $\pm 1\%$ input over $M = 200$).

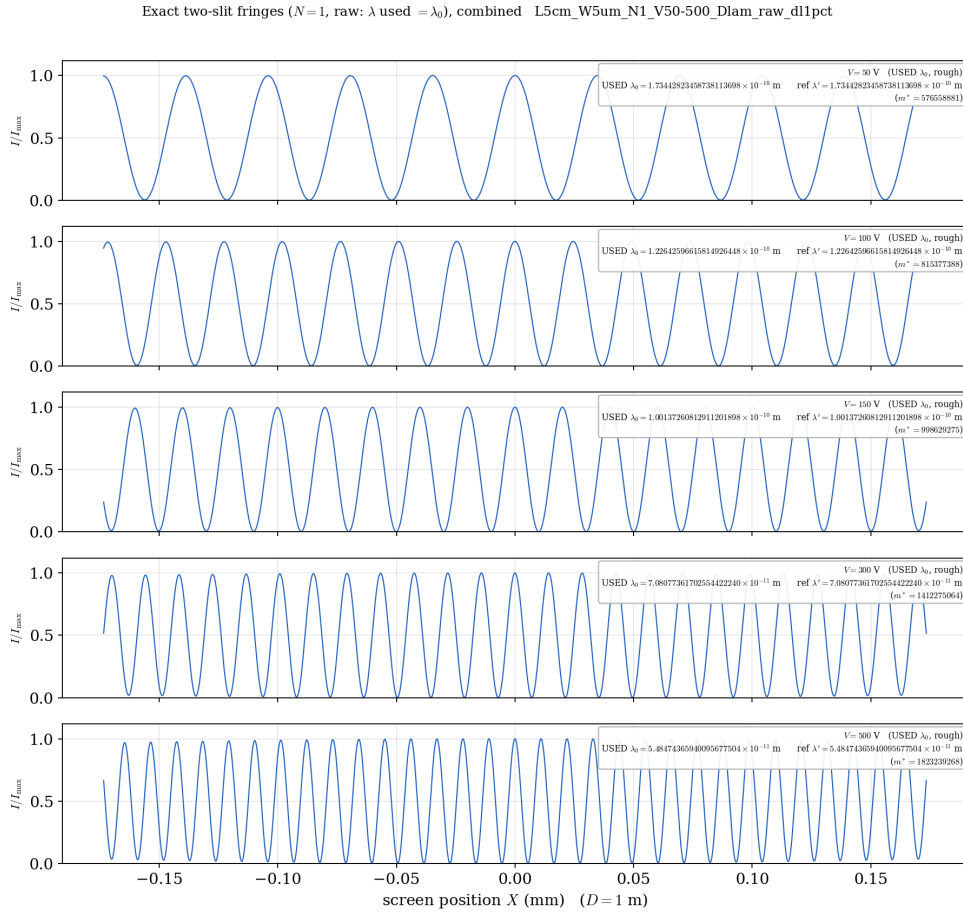


Figure 2: $N = 1$ accumulated fringes ($V = 50, 100, 150, 300, 500$ V, $D = 1$ m, common window). Each row is the exact re-computed accumulation of 200 electrons with source position fluctuating over $\pm\lambda_0/2$ and wavelength over $\pm 1\%$. Higher voltages give denser fringes ($\Delta X = 0.035 \rightarrow 0.011$ mm), and the $\pm 1\%$ wavelength spread slightly lowers the visibility toward the window edges (§3).

5 4. $N > 1$: survival of interference, and fundamental-wavelength invariance of the observed wavelength

We compute the multi-harmonic cases $N = 9, 17$ (Table 1).

Uncorrected (raw, using λ'_0 directly): the inter-harmonic absolute alignment phase ($r_{\text{avg}}/\lambda \sim 5 \times 10^8$) is generally not aligned, the localization collapses, and multiple split peaks appear. Because of the multiple peaks **a unique period is not defined and the wavelength cannot be read from the fringe spacing**. The slope/ h obtained by formally applying the same readout is 0.139 for $N = 9$ and 0.081 for $N = 17$, but **these are diagnostics for a collapsed pattern with no stable fringes and do not represent a valid wavelength measurement**.

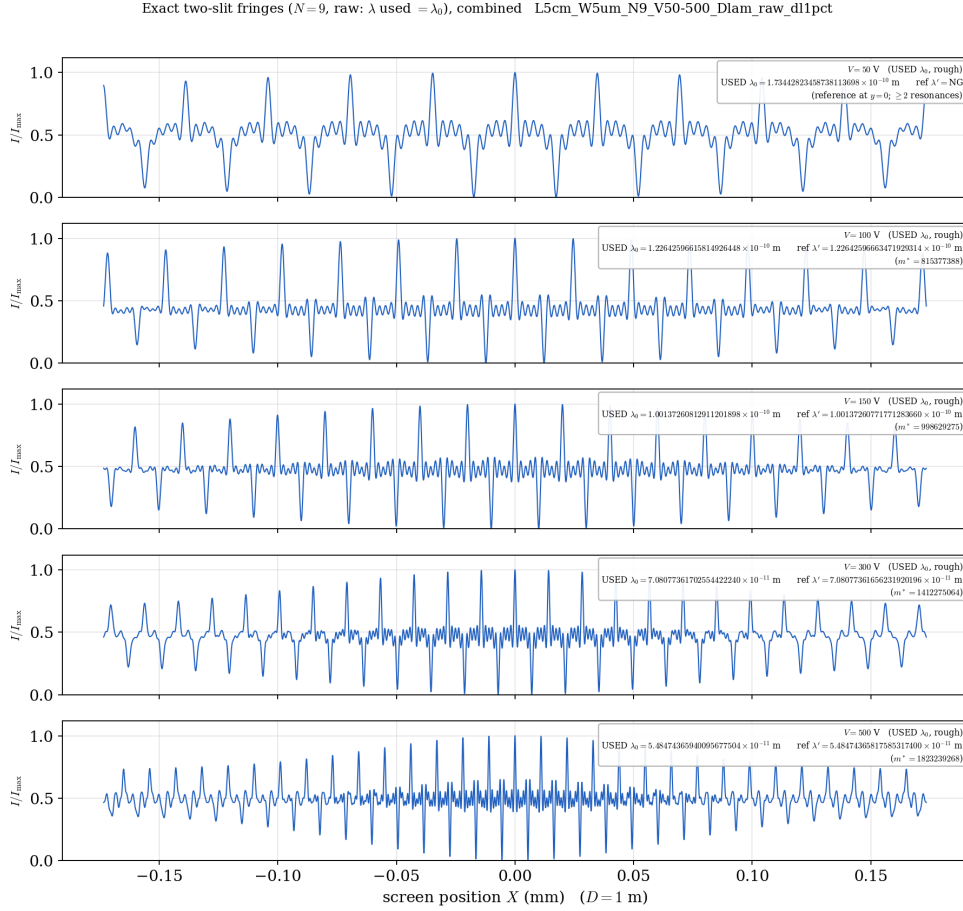


Figure 3: $N =$

9 **uncorrected** (raw) accumulated fringes. Each period is not a single sharp peak but **split multiple peaks (comb-like)**, and the center $s = 0$ is not a main peak = stable interference does not hold. In this state a unique period cannot be read.

Corrected (align, using λ'): each electron's λ'_0 is aligned to the nearest resonance $\lambda' = 2r_{\text{avg}}/M$ of its geometry $r_{\text{avg}}(y)$ (= the survival filter, §2.5). All harmonic phases align and a train of sharp single localized peaks is restored (Figure 4). Reading the wavelength geometrically from the main-peak spacing ΔX of this sharp train, via $\lambda = \Delta X \cdot W/D$, agrees with the λ' obtained through A (the wavelength that could not be read in raw is recovered in align). The slope of $\langle p\lambda' \rangle$ is 0.99973 for $N = 9$ and 0.99972 for $N = 17$.

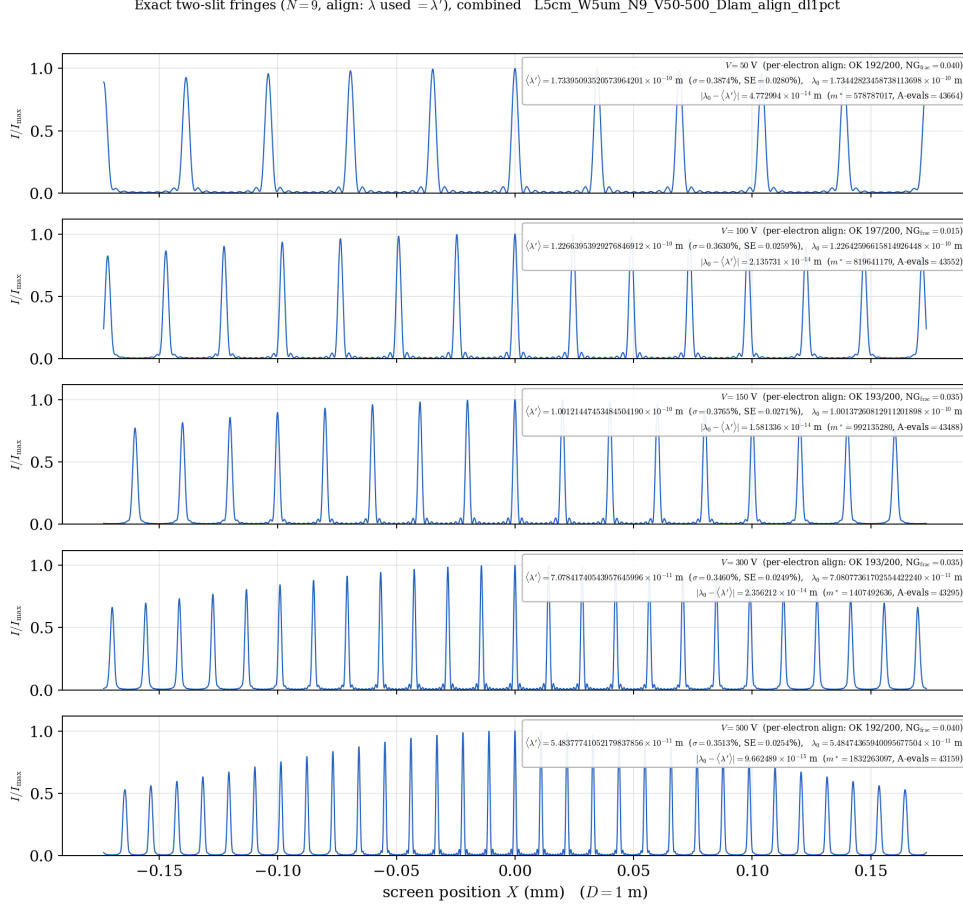


Figure 4: $N =$

9 aligned (align) accumulated fringes. The comb-like split of Figure 3 turns, via the survival filter, into **sharp single localized peaks (with the S_N side-lobes)**. Each panel annotates $n_{OK}/200$, the exclusion rate, $\langle \lambda' \rangle$ (20 digits), $\sigma\%$, SE%. λ' coincides with λ_0 to relative $\sim 10^{-9}$; the alignment is a tiny snap to a geometric resonance.

Exact two-slit fringes ($N=17$, align: λ used = λ'), combined L5cm_W5um_N17_V50-500_Dlam_align_d11pct

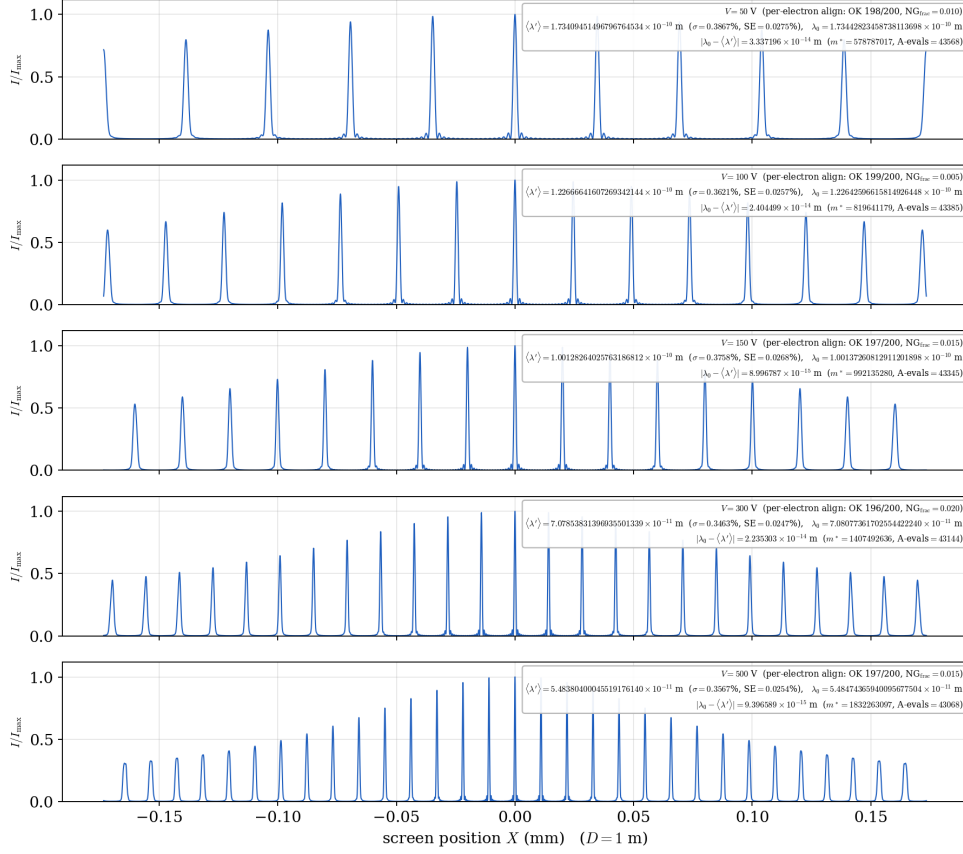


Figure 5: $N =$

17 aligned. With 9 harmonics the localization is even sharper. **The peak positions and spacing are invariant with respect to N ; N controls only the resolution (sharpness $\sim 1/(N+1)$).**

N	# harmonics K	raw slope/ h (diagnostic)	align slope/ h	localization width $\sim 1/(N+1)$
1	1	0.99977	0.99977	sinusoid (no localization)
9	5	0.139 (collapsed, invalid)	0.99973	$\sim 1/10$
17	9	0.081 (collapsed, invalid)	0.99972	$\sim 1/18$

Table 1: N dependence of slope/ h ($\pm 1\%$ wavelength fluctuation, 200-electron average). The raw values are formal readouts of a collapsed pattern and are not wavelength measurements. When aligned, slope/ $h \approx 1$ regardless of N .

After alignment, the main-peak positions on the screen are given by $Ws/\lambda' \in \mathbb{Z}$, so the main-peak spacing in the paraxial coordinate $X \simeq Ds$ is $\Delta X = \lambda'D/W$. The odd-harmonic count N changes the sharpness and the side-lobe structure, but this **main-peak spacing is set by the fundamental wavelength λ' alone and is independent of N .**

Central result of this section (a structural result independent of the origin of λ_0): increasing N , the spacing of the sharp localized-peak train after align (= the central fringe spacing $\Delta X = \lambda_0 D/W$) is set by the fundamental wavelength λ_0 and is independent of the harmonic count N . That is, even when finer internal structure (harmonics) than λ_0 is packed in, the wavelength scale observed under two-slit geometry is **invariant** as the fundamental

wavelength, and what N changes is only the sharpness of localization (resolution). The trade-off **high N = high resolution but high fragility (tolerance band $\sim 1/(2N)$)** could be compared in future work with the resolution–stability trade-off of atomic clocks and electron interferometers (this paper makes no detailed comparison) [1].

6 5. p vs $1/\lambda$ and the simulated observation of $p\lambda$ constancy

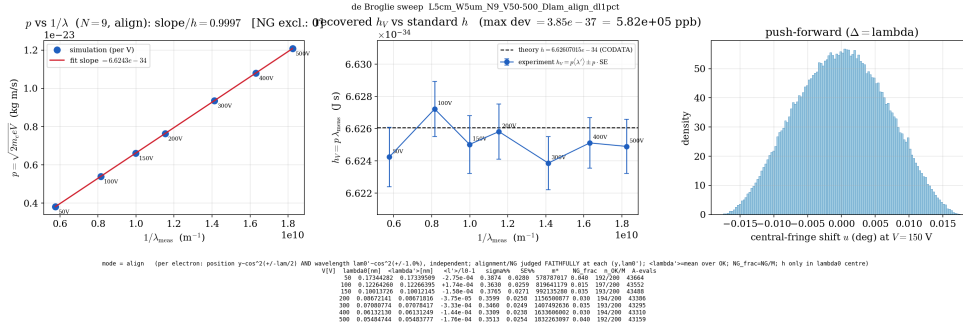


Figure 6: Sum-

mary of $N = 9$ align ($\pm 1\%$ wavelength fluctuation, 200 electrons per voltage). **How to read (three panels):** **Left** = p vs $1/\langle\lambda'\rangle$ (a straight line, slope = h). The linearity is a visualization of the relation, not a measure of precision. **Center** = the per-voltage $h_V = p\langle\lambda'\rangle$ plotted with $\pm p$ -SE error bars, extremely zoomed around CODATA h (dashed, at the center). All error bars contain h , so $\langle p\lambda'\rangle$ is statistically consistent with h (scatter $\sim 0.03\%$ = the 200-average of the $\pm 1\%$ input). **Right** = the distribution of the central-fringe positional shift caused by the source-position fluctuation ($\cos^2$). The bottom table gives, per voltage, $\langle\lambda'\rangle$, $\sigma\%$, SE%, exclusion rate, $n_{\text{OK}}/200$, and the number of A evaluations. Representative values: $\sigma \approx 0.35\text{--}0.39\%$ (matching the theoretical 0.362%), $\text{SE} \approx 0.025\text{--}0.028\%$, exclusion rate $1.5\text{--}4.0\%$, $n_{\text{OK}} = 192\text{--}197/200$.

6.1 5.1 Explicit statement of self-consistency

We use $\lambda_0 = h/p$ as the **center** of each electron’s initial wavelength λ'_0 . h enters at this single place and returns as $p\lambda' \approx h$ through $\lambda' \approx \lambda_0$ (snap $\sim 10^{-9}$). Hence “recovering h ” is itself **self-consistent (circular)** and is not an independent numerical determination of h . The $\pm 1\%$ wavelength fluctuation and the 200-electron average are a realism device against misreading; the many-electron average $\langle p\lambda'\rangle \rightarrow h$ is a consequence of the cancellation due to λ'_0 being symmetric about the center $\lambda_0 = h/p$ (interference does not independently select or correct the true value h). The **survival of the structure and the fundamental-wavelength invariance** discussed in §4 are separate results that do not depend on this circularity.

6.2 5.2 What this paper does not derive (scope of the claim)

(a) **Not a first-principles derivation of $p\lambda = h$.** $p = \sqrt{2m_e e V}$ is the classical momentum set by the accelerating voltage (no h), and λ is the representative wavelength $\lambda_0 = h/p$ given to the source and the effective wavelength λ' extracted from interference. Since λ_0 itself is given using h , $p\lambda' \approx h$ is a self-consistent result and is not a derivation that h emerges from the conjugacy of p and λ .

(b) **Not a derivation of a $\Delta x \Delta p \sim h$ type uncertainty relation.** What the localized odd-harmonic wave of this paper essentially possesses is only the **classical (Fourier) relation** $\Delta x \Delta k \sim 1$ between the localization width Δx and the wavenumber width Δk . To connect this to a momentum width and obtain $\Delta x \Delta p \sim \hbar$ requires, as a separate physical input, $p = \hbar k$ (= the de Broglie correspondence itself). This paper is a stage that assumes it, and does not derive the uncertainty relation.

Therefore what this paper shows is a classical-wave-model result: **even when finer localized odd-harmonic structure than λ_0 is given to a wave whose representative wavelength is at the de Broglie scale, it survives stably as sharp interference under two-slit geometry and alignment, and the simulated-observed central fringe spacing is invariant as the fundamental wavelength λ_0** —it is neither a derivation of the value h nor a derivation of conjugacy / uncertainty.

6.3 5.3 Correspondence with real experiments (a limited analogy)

Davisson–Germer and other real matter-wave experiments also narrow the wavelength range by coarse estimation and then increase precision by lattice diffraction, but in real experiments the diffraction angle, an external measured datum, supplies λ independently. This model generates fringes from $\lambda_0 = h/p$ and reads them back, so it has no independent measurement path and is not isomorphic in this respect. The exclusion (NG) and per-electron alignment are not a model of quantum measurement or wave-packet collapse but a classical filtering operation of “adopting only trials in which stable interference held”, and can be positioned as one concrete example of the quantum–classical correspondence (§2.5, §7).

7 6. Limits of the computation and the model

- **Self-reference:** since the center of λ'_0 is h/p , the numerical value of h is not determined independently.
- **Triviality from lattice density:** since the resonance-lattice spacing $\sim 10^{-19}$ m is dense, $\lambda' \approx \lambda_0$ is mathematically almost inevitable. The claim of this paper is placed not on “recovering a value” but on “interference survival of the localized structure and fundamental-wavelength invariance” (§4, §7). Computing in a coarse-lattice regime (drastically smaller L , or an analogous experiment at optical wavelengths) would make the deviation between λ' and λ_0 visible and highlight the non-triviality of the mechanism (future work).
- **Numerical precision:** since $r_{\text{avg}}/\lambda \sim 10^9$, the alignment phase for $N > 1$ is at the float64 limit; $A(\lambda) = I(0)/4K^2$ (absolute alignment) is computed and $\lambda' = 2r_{\text{avg}}/M$ is constructed directly to avoid cancellation.
- **Non-countability of the order M :** $M \sim 10^9$ is unlike the small orders $n = 1, 2, 3$ of Bragg diffraction and cannot be identified by counting fringes, so it is not fully isomorphic to crystal Bragg diffraction.
- **Smallness of the push-forward:** the fringe shift for the source-position fluctuation $\pm\lambda_0/2$ (a basic assumption of this paper) is $\sim 0.02^\circ$, small because of paraxiality, and its contribution to the central wavelength scale is small (a correct consequence under the assumption).
- **Representative values:** L, W, V, Δ, δ are representative values, so the quantitative precision is limited.

8 7. Conclusion and outlook

In a single-source double-slit classical wave model, we gave a localized odd-harmonic wave ($N > 1$) with a de Broglie-scale representative wavelength $\lambda_0 = h/p$, and fluctuated each electron’s position ($\pm\lambda_0/2$) and wavelength ($\pm 1\%$) independently to compute the interference, alignment, and exclusion exactly. $N = 1$ is a check of the procedure. **The central result of this paper is not the value of h , but (i) that a classical wave with structure finer than λ_0**

can survive stably as a sharp single-peak interference under two-slit geometry, and (ii) that the central spacing of the surviving fringes remains at the fundamental wavelength λ_0 , invariant with respect to the harmonic count N (N controls only the resolution $\sim 1/(N+1)$). Under a symmetric fluctuation centered at $\lambda_0 = h/p$, $\langle p\lambda' \rangle$ agrees with h within the standard error, but this is a self-consistency due to lattice density and not an independent derivation.

This paper is neither a first-principles derivation of $p\lambda = h$ nor a derivation of a $\Delta x \Delta p \sim h$ type uncertainty relation. The value of this paper is not that it “derived h ”, but that it verified, in an explicit classical wave model, that even with localized structure finer than the de Broglie wavelength, the observed interference wavelength remains at the de Broglie scale (fundamental-wavelength invariance).

Outlook: (a) visualize $\lambda' \neq \lambda_0$ in a coarse-lattice regime (small L / optical wavelengths) to show the non-triviality of the mechanism, (b) quantify the N - and fluctuation-width- δ -dependence of the exclusion rate, (c) position the alignment sensitivity (high resolution and high fragility) as a universal structure shared with atomic clocks and electron interferometers, (d) publish an English version.

9 References

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- [3] L. de Broglie, “Recherches sur la théorie des quanta,” *Annales de Physique* **10**(3), 22–128 (1925). DOI: 10.1051/anphys/192510030022. (The formal journal information / DOI should be verified.)
- [4] M. Born, “Zur Quantenmechanik der Stoßvorgänge,” *Zeitschrift für Physik* **37**, 863–867 (1926). DOI: 10.1007/BF01397477. — In this paper the $\cos^2$ probability density used for the source-position distribution is referenced as an assumption of a **Born-type** ($|\psi|^2$ -type) **probability distribution** (§2.4). It does not derive or apply the Born rule itself.

Note: the numerics and figures were generated by the parameterized `debroglie_plambda_sweep.py` (exact two-slit intensity + independent $\cos^2$ fluctuation of position and wavelength + per-electron accumulation) and the absolute-alignment A search solver `debroglie_align_lambda.py`. The generation condition of each figure is encoded in its filename (e.g. `..._N9_V50-500_Dlam_align_d11pct` = $N=9$, aligned, wavelength fluctuation $\pm 1\%$).